

BOARD ASSESSMENT SERIES 2026

10 SLO-BASED MODEL PAPERS WITH SOLUTIONS

Subject: Chemistry (Class 9 / SSC-I)

Exam Pattern: 85-Marks FBISE Standards

Chapters: National Book Foundation National Curriculum Model

COMPREHENSIVE ASSESSMENT	Covers entire curriculum from Units 1 to 19, including practical experimental skills.
85-MARKS EXAM PATTERN	Conforms strictly to FBISE: 17 MCQs, 14 short answer pairs, and 4 long extensive answer pairs.
100% EXPLICIT SOLUTIONS	Every question (including all 'OR' choices) is fully solved with precise chemical equations and grading splits.
SLO DIRECTED DESIGN	Aligned with Student Learning Objectives (SLOs) from the National Book Foundation (NBF) Textbook.

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Note on FBISE assessment guidelines: This textbook and assessment model strictly evaluate Student Learning Outcomes (SLOs) rather than rote-learning of exercise questions. Questions have been formulated based on analytical reasoning, conceptual knowledge, and practical applications as guided by the Federal Board of Intermediate and Secondary Education (FBISE), Islamabad. All standard chemical formulas are represented accurately.

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 1

Syllabus Focus: Units 1 & 2 (Nature of Science & Matter)

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. Chemistry deals with the composition, structure, properties, and changes of:

- A) Forces
- B) Energy
- C) Matter
- D) Motion

Q2. Which of the following branch deals with carbon-containing compounds except carbonates?

- A) Inorganic
- B) Organic
- C) Biochemistry
- D) Analytical

Q3. Physical chemistry is primarily concerned with the relationship between:

- A) Carbon compounds
- B) Biomolecules
- C) Laws and theories of matter
- D) Industrial products

Q4. Which sub-field of chemistry focuses on chemical processes in living organisms?

- A) Environmental
- B) Biochemistry
- C) Nuclear
- D) Analytical

Q5. The study of chemical techniques for water purification represents:

- A) Astrochemistry
- B) Medicinal chemistry
- C) Environmental chemistry
- D) Nuclear chemistry

Q6. Bose-Einstein Condensate (BEC) represents which state of matter?

- A) Solid
- B) Liquid
- C) Fifth
- D) Fourth (Plasma)

Q7. Which state of matter has low density at normal conditions due to large spaces?

- A) Solid
- B) Liquid
- C) Gas
- D) Crystal

Q8. What is the crystalline allotrope of carbon with a 2D hexagonal layer structure?

- A) Diamond
- B) Graphite
- C) Buckyballs
- D) Amorphous carbon

Q9. Which allotrope of carbon has a soccer-ball-like fused hollow ring structure?

- | | |
|--------------------|----------------------|
| A) Diamond | B) Graphite |
| C) Carbon nanotube | D) Buckyballs (C-60) |

Q10. A mixture of two or more substances that is completely homogeneous is called a:

- | | |
|---------------|-------------|
| A) Suspension | B) Solution |
| C) Colloid | D) Compound |

Q11. A solution containing the maximum amount of solute at a given temperature is:

- | | |
|-------------------|----------------|
| A) Saturated | B) Unsaturated |
| C) Supersaturated | D) Dilute |

Q12. If a solution can dissolve more solute at a given temperature, it is called:

- | | |
|----------------|-------------------|
| A) Saturated | B) Supersaturated |
| C) Unsaturated | D) Concentrated |

Q13. A mixture whose particles range between 1 and 1000 nm in diameter is a:

- | | |
|---------------|-------------|
| A) Solution | B) Colloid |
| C) Suspension | D) Compound |

Q14. Which of the following is a classic example of a suspension?

- | | |
|----------------|----------|
| A) Sea water | B) Milk |
| C) Muddy water | D) Brass |

Q15. Macroscopic properties like density, fluidity, and compressibility help distinguish:

- | | |
|---------------------|-------------|
| A) Isotopes | B) Elements |
| C) States of matter | D) Cations |

Q16. What distinguishes a compound from a mixture?

- | | |
|-----------------------|-----------------------------------|
| A) Physical combining | B) Chemical combining of elements |
| C) Homogeneous always | D) Easily separated |

Q17. The process of applying scientific knowledge to practical applications is called:

- | | |
|----------------|--------------|
| A) Engineering | B) Science |
| C) Technology | D) Chemistry |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Define Organic Chemistry and differentiate it from Inorganic Chemistry.

OR

Option B: Define Biochemistry and state its primary focus in living organisms.

Question 2 (part 2).

Option A: What is Green Chemistry, and why is it important for the environment?

OR

Option B: Describe Environmental Chemistry and its role in combating pollution.

Question 2 (part 3).

Option A: Compare the macroscopic properties of gases and solids regarding density and compressibility.

OR

Option B: Explain why gases are highly compressible while solids are virtually incompressible.

Question 2 (part 4).

Option A: Define an Element and describe how it differs from a Compound.

OR

Option B: Define a Mixture and distinguish between Homogeneous and Heterogeneous mixtures.

Question 2 (part 5).

Option A: What are Allotropes? Name three allotropic forms of carbon.

OR

Option B: Describe the structure and electrical conductivity of Graphite.

Question 2 (part 6).

Option A: Explain why Diamond is extremely hard and acts as an electrical insulator.

OR

Option B: Describe Buckyballs (C-60) and state their physical properties and molecular shape.

Question 2 (part 7).

Option A: Define a Solution and explain the terms Solute and Solvent with examples.

OR

Option B: What is an Aqueous Solution? Give two common laboratory examples.

Question 2 (part 8).

Option A: Differentiate between Saturated and Unsaturated solutions based on solute concentration.

OR

Option B: What is a Supersaturated Solution? How can it be prepared in the laboratory?

Question 2 (part 9).

Option A: State the effect of temperature on the solubility of NaCl and Ca(OH)₂.

OR

Option B: Describe how temperature affects the solubility of NH₄Cl compared to Ce₂(SO₄)₃.

Question 2 (part 10).

Option A: Define Colloids and explain the Tyndall Effect with a real-life example.

OR

Option B: What is a Suspension? Explain why it is heterogeneous and unstable.

Question 2 (part 11).

Option A: Differentiate between 'Science', 'Technology', and 'Engineering'.

OR

Option B: Provide a real-life example of how Science, Technology, and Engineering collaborate.

Question 2 (part 12).

Option A: How does Analytical Chemistry assist forensic scientists in criminal investigations?

OR

Option B: Explain the role of Medicinal Chemistry in modern healthcare and pharmacology.

Question 2 (part 13).

Option A: Explain the concept of Plasma as the fourth state of matter.

OR

Option B: What is a Bose-Einstein Condensate, and under what conditions does it exist?

Question 2 (part 14).

Option A: What is an alloy? Give the composition and uses of Brass.

OR

Option B: Differentiate between concentrated and dilute solutions.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Analyze the three main allotropes of carbon: Diamond, Graphite, and Buckyballs. Differentiate them based on hybridization/structure, conductivity, and hardness. *[Marks: 6.5]*

OR

Option B: Explain the classification of matter. Draw the complete hierarchy diagram showing pure substances, mixtures, elements, compounds, colloids, and suspensions. *[Marks: 6.5]*

Question 4.

Option A: Describe the preparation and testing of Saturated, Unsaturated, and Supersaturated solutions using sugar and water as an experimental example. *[Marks: 6.5]*

OR

Option B: Define Solubility. Analyze the effect of temperature on solubility curves, citing examples of solute types whose solubility increases, decreases, or remains constant. *[Marks: 6.5]*

Question 5.

Option A: Compare Solutions, Colloids, and Suspensions in a comprehensive matrix based on particle size, homogeneity, Tyndall effect, filtration, and settling behavior. *[Marks: 6.5]*

OR

Option B: Explain with examples how science, technology, and engineering are interconnected. Discuss three daily life applications of chemistry. *[Marks: 6.5]*

Question 6.

Option A: Formulate essential questions for physical, organic, inorganic, and biochemistry. Explain how these fields solve global challenges like energy and health. *[Marks: 6.5]*

OR

Option B: Discuss the characteristics of water-filtration and solar-energy systems, detailing the role of chemical principles in their design. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 1

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Chemistry is defined as the science that investigates the materials of the universe (matter) and their changes [1].
Q2	B	Organic Chemistry deals with substances containing carbon, excluding carbonates, bicarbonates, and oxides [1].
Q3	C	Physical chemistry is the branch that deals with laws and theories to understand structure and changes of matter [1].
Q4	B	The branch that deals with physical and chemical changes in living organisms is called biochemistry [1].
Q5	C	Environmental chemistry deals with the study of chemical and toxic substances that pollute the environment and mitigation [1].
Q6	C	BEC is defined as the state of matter where separate atoms are cooled to near absolute zero [16].
Q7	C	Gases have low density at normal conditions due to large spaces between molecules [17].
Q8	B	Graphite is composed of flat 2D layers of hexagonally arranged carbon atoms [18].
Q9	D	Buckyballs (C-60) have a football-like fused hollow ring structure [19].
Q10	B	A solution is a homogeneous mixture of two or more substances [19].
Q11	A	A solution containing maximum amount of solute at a given temperature is saturated [21].
Q12	C	A solution containing lesser amount of solute than required to saturate is unsaturated [21].
Q13	B	A colloid has particle sizes between 1 and 1000 nm [24].
Q14	C	Muddy water is an example of suspension, which is heterogeneous and separates [24].
Q15	C	These macroscopic properties visualize the differences in states of matter [17].
Q16	B	A compound is formed when two or more elements chemically combine [18].
Q17	C	Technology is the process of applying scientific knowledge to practical applications [10].

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Define Organic Chemistry and differentiate it from Inorganic Chemistry.

Model Answer A: Organic Chemistry deals with covalent carbon compounds. Inorganic Chemistry deals with all other elements and compounds [1].

Option B: Define Biochemistry and state its primary focus in living organisms.

Model Answer B: Biochemistry deals with the chemical substances and physical/chemical changes in living organisms [1].

Question 2 (part 2) Solution:

Option A: What is Green Chemistry, and why is it important for the environment?

Model Answer A: Green Chemistry focuses on designing chemical products and processes that reduce or eliminate hazardous substances [1].

Option B: Describe Environmental Chemistry and its role in combating pollution.

Model Answer B: Environmental Chemistry deals with chemical and toxic substances in the environment and their biological impacts [1].

Question 2 (part 3) Solution:

Option A: Compare the macroscopic properties of gases and solids regarding density and compressibility.

Model Answer A: Gases have low density and high compressibility; solids have high density and are incompressible [17].

Option B: Explain why gases are highly compressible while solids are virtually incompressible.

Model Answer B: Gases have large empty spaces between molecules, allowing compression; solid molecules are tightly packed [17].

Question 2 (part 4) Solution:

Option A: Define an Element and describe how it differs from a Compound.

Model Answer A: An element consists of only one type of atom. A compound consists of different elements chemically combined [18].

Option B: Define a Mixture and distinguish between Homogeneous and Heterogeneous mixtures.

Model Answer B: A mixture is physically combined and can be homogeneous (uniform, e.g., solution) or heterogeneous (non-uniform, e.g., suspension) [19].

Question 2 (part 5) Solution:

Option A: What are Allotropes? Name three allotropic forms of carbon.

Model Answer A: Allotropes are different physical forms of the same element. Carbon allotropes: Diamond, Graphite, Buckyballs [18].

Option B: Describe the structure and electrical conductivity of Graphite.

Model Answer B: Graphite has 2D hexagonal layers with free delocalized electrons, making it a good electrical conductor [18].

Question 2 (part 6) Solution:

Option A: Explain why Diamond is extremely hard and acts as an electrical insulator.

Model Answer A: Diamond has a rigid 3D tetrahedral network of covalently bonded carbon with no free electrons, making it hard and an insulator [18, 19].

Option B: Describe Buckyballs (C-60) and state their physical properties and molecular shape.

Model Answer B: Buckyballs are football-shaped hollow cages of 60 carbon atoms, containing pentagons and hexagons [19].

Question 2 (part 7) Solution:

Option A: Define a Solution and explain the terms Solute and Solvent with examples.

Model Answer A: A solution is a homogeneous mixture. Solute is dissolved (e.g., salt); solvent does the dissolving (e.g., water) [19].

Option B: What is an Aqueous Solution? Give two common laboratory examples.

Model Answer B: An aqueous solution uses water as the solvent. Examples: NaCl in water, CuSO₄ in water [20].

Question 2 (part 8) Solution:

Option A: Differentiate between Saturated and Unsaturated solutions based on solute concentration.

Model Answer A: Saturated has max solute at that temperature; unsaturated has less than saturated capacity [21].

Option B: What is a Supersaturated Solution? How can it be prepared in the laboratory?

Model Answer B: Supersaturated contains more solute than saturated. Prepared by heating saturated solution, adding solute, and cooling slowly [21].

Question 2 (part 9) Solution:

Option A: State the effect of temperature on the solubility of NaCl and Ca(OH)₂.

Model Answer A: NaCl solubility increases slightly with temperature; Ca(OH)₂ solubility decreases with increasing temperature [22].

Option B: Describe how temperature affects the solubility of NH₄Cl compared to Ce₂(SO₄)₃.

Model Answer B: NH₄Cl solubility increases sharply with temperature, while Ce₂(SO₄)₃ decreases [22].

Question 2 (part 10) Solution:

Option A: Define Colloids and explain the Tyndall Effect with a real-life example.

Model Answer A: Colloids have intermediate particle size and scatter light (Tyndall Effect), e.g., light beam through milk [24].

Option B: What is a Suspension? Explain why it is heterogeneous and unstable.

Model Answer B: A suspension has large particles (>1000 nm) that do not dissolve and settle down upon standing, e.g., muddy water [24].

Question 2 (part 11) Solution:

Option A: Differentiate between 'Science', 'Technology', and 'Engineering'.

Model Answer A: Science seeks to understand nature. Technology applies science. Engineering designs and builds systems [9, 10].

Option B: Provide a real-life example of how Science, Technology, and Engineering collaborate.

Model Answer B: Harnessing solar energy: Science studies photovoltaic effect, Technology creates solar panels, Engineering constructs solar farms [11].

Question 2 (part 12) Solution:

Option A: How does Analytical Chemistry assist forensic scientists in criminal investigations?

Model Answer A: Forensic chemistry uses analytical tools to identify physical traces, drugs, toxins, and paint at crime scenes [1].

Option B: Explain the role of Medicinal Chemistry in modern healthcare and pharmacology.

Model Answer B: Medicinal chemistry designs and synthesizes drugs that interact with biological targets to treat diseases [1].

Question 2 (part 13) Solution:

Option A: Explain the concept of Plasma as the fourth state of matter.

Model Answer A: Plasma is highly ionized gas consisting of free electrons, positive ions, and excited neutral atoms [16, 17].

Option B: What is a Bose-Einstein Condensate, and under what conditions does it exist?

Model Answer B: BEC is a state of matter formed when separate atoms are cooled to temperatures very close to absolute zero [16].

Question 2 (part 14) Solution:

Option A: What is an alloy? Give the composition and uses of Brass.

Model Answer A: An alloy is a solid solution of metals. Brass is made of copper and zinc, used in decorative items and plumbing [20].

Option B: Differentiate between concentrated and dilute solutions.

Model Answer B: Concentrated solution contains relatively large solute; dilute contains a relatively small amount [22].

Section C Model Answers

Question 3 Solution:

Option A: Analyze the three main allotropes of carbon: Diamond, Graphite, and Buckyballs. Differentiate them based on hybridization/structure, conductivity, and hardness.

Detailed Model Answer A: Diamond has tetrahedral structure, no free electrons (insulator), and is extremely hard. Graphite has planar hexagonal sheets, free electrons (conductor), and is soft/slippery. Buckyballs have hollow spherical cages [18, 19].

Option B: Explain the classification of matter. Draw the complete hierarchy diagram showing pure substances, mixtures, elements, compounds, colloids, and suspensions.

Detailed Model Answer B: Matter divides into Pure Substances (Elements & Compounds) and Mixtures (Homogeneous like Solutions, Heterogeneous like Colloids and Suspensions) [19].

Question 4 Solution:

Option A: Describe the preparation and testing of Saturated, Unsaturated, and Supersaturated solutions using sugar and water as an experimental example.

Detailed Model Answer A: Add sugar to 100g water until it stops dissolving (saturated). Add more water (unsaturated). Heat saturated solution, add more sugar, and cool slowly (supersaturated) [21].

Option B: Define Solubility. Analyze the effect of temperature on solubility curves, citing examples of solute types whose solubility increases, decreases, or remains constant.

Detailed Model Answer B: Solubility is max solute in 100g solvent. $\text{KNO}_3/\text{NH}_4\text{Cl}$ increase with temperature. $\text{Ca}(\text{OH})_2/\text{SO}_2$ gas decrease with temperature. NaCl remains nearly constant [22, 23].

Question 5 Solution:

Option A: Compare Solutions, Colloids, and Suspensions in a comprehensive matrix based on particle size, homogeneity, Tyndall effect, filtration, and settling behavior.

Detailed Model Answer A: Solutions ($<1\text{nm}$, homogeneous, no Tyndall, passes filter, doesn't settle). Colloids (1-1000nm, heterogeneous-looking homogeneous, Tyndall, passes filter, doesn't settle). Suspensions ($>1000\text{nm}$, heterogeneous, Tyndall, retained, settles) [24].

Option B: Explain with examples how science, technology, and engineering are interconnected. Discuss three daily life applications of chemistry.

Detailed Model Answer B: Science discovers laws; engineering designs systems; technology manufactures. Chemistry applications: medicines (organic), batteries (inorganic), forensic tests (analytical) [8, 9, 10].

Question 6 Solution:

Option A: Formulate essential questions for physical, organic, inorganic, and biochemistry. Explain how these fields solve global challenges like energy and health.

Detailed Model Answer A: Physical: How do bonds form? Organic: Why is carbon the backbone? Inorganic: How does periodic table organize elements? Biochemistry: How do proteins function? These address energy (batteries) and health (drugs) [7, 8].

Option B: Discuss the characteristics of water-filtration and solar-energy systems, detailing the role of chemical principles in their design.

Detailed Model Answer B: Solar-energy uses semiconductor chemistry. Water filtration uses chemical coagulation, sedimentation, and active carbon adsorption to remove contaminants [11].

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 2

Syllabus Focus: Units 3 & 5 (Atomic Structure & Chemical Bonding)

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. In Rutherford's experiment, alpha particles were bombarded on a thin foil of:

- A) Silver
- B) Gold
- C) Copper
- D) Gold-plated Iron

Q2. According to Bohr's model, the angular momentum of an electron is an integral multiple of:

- A) $h/2$
- B) $h/2\pi$
- C) $2\pi/h$
- D) h/λ

Q3. The maximum number of electrons that a d-subshell can accommodate is:

- A) 2
- B) 6
- C) 10
- D) 14

Q4. Which isotope is commonly used to irradiate cancer cells to shrink tumors?

- A) Carbon-14
- B) Iodine-131
- C) Cobalt-60
- D) Sodium-24

Q5. Atomic number or proton number is indicated by the symbol:

- A) A
- B) Z
- C) N
- D) M

Q6. Which bond is formed by the complete transfer of electrons from one atom to another?

- A) Covalent
- B) Ionic
- C) Coordinate Covalent
- D) Metallic

Q7. An atom is electrically neutral because:

- A) No neutrons
- B) Protons equal Neutrons
- C) Protons equal Electrons
- D) Nucleus is neutral

Q8. The relative mass of an electron is approximately:

- A) 1 amu
- B) 1836 amu
- C) $1/1836$ amu
- D) 0.5 amu

Q9. Which sub-shell has the lowest energy and is filled first?

- | | |
|-------|-------|
| A) 2s | B) 1s |
| C) 2p | D) 3s |

Q10. An ionic bond is formed between sodium (Na) and chlorine (Cl). Na becomes:

- | | |
|------------|-------------|
| A) Anion | B) Cation |
| C) Isotope | D) Molecule |

Q11. Which of the following compounds has a coordinate covalent bond?

- | | |
|---------------------------------|--------------------|
| A) H ₂ O | B) NH ₃ |
| C) NH ₄ ⁺ | D) CH ₄ |

Q12. The electronegativity difference in a polar covalent bond is:

- | | |
|----------------------------------|---------------------|
| A) Zero | B) Less than 0.4 |
| C) Significant but less than 1.7 | D) Greater than 1.7 |

Q13. In a metallic bond, metal cations are held together by a:

- | | |
|------------------------------------|-----------------------------|
| A) Sea of delocalized electrons | B) Shared pair of electrons |
| C) Electrostatic force with anions | D) Hydrogen bonds |

Q14. The relative atomic mass scale is based on which standard isotope?

- | | |
|---------------|----------------|
| A) Hydrogen-1 | B) Oxygen-16 |
| C) Carbon-12 | D) Uranium-238 |

Q15. Chlorine-35 and Chlorine-37 are examples of:

- | | |
|---------------|------------|
| A) Allotropes | B) Isobars |
| C) Isotopes | D) Cations |

Q16. Which intermolecular force is responsible for the high boiling point of water?

- | | |
|-------------------------|----------------------|
| A) Dipole-dipole forces | B) London dispersion |
| C) Hydrogen bonding | D) Covalent bonding |

Q17. According to octet rule, atoms try to acquire how many electrons in valence shell?

- | | |
|-------|-------|
| A) 2 | B) 8 |
| C) 18 | D) 32 |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Differentiate between Rutherford's Atomic Model and Bohr's Atomic Theory regarding electron orbits.

OR

Option B: Describe the defects of Rutherford's Atomic Model and how Bohr resolved them.

Question 2 (part 2).

Option A: Define isotopes and list the isotopes of hydrogen with their proton and neutron counts.

OR

Option B: Describe three practical applications of radioactive isotopes in medicine and industry.

Question 2 (part 3).

Option A: Define Aufbau Principle and write the electronic configuration of Sodium ($Z=11$) and Chlorine ($Z=17$).

OR

Option B: Differentiate between Shells and Sub-shells. List the sub-shells of the M shell.

Question 2 (part 4).

Option A: Define Octet and Duplet rules. Give one example of an element obeying each.

OR

Option B: Explain why noble gases are highly stable and chemically inert.

Question 2 (part 5).

Option A: Describe the formation of Cations and Anions with one chemical equation for each.

OR

Option B: Write the electronic configuration of Magnesium ion (Mg^{2+}) and Oxide ion (O^{2-}).

Question 2 (part 6).

Option A: Define Ionic Bond. Illustrate the formation of sodium chloride (NaCl) using Lewis dot structure.

OR

Option B: Define Covalent Bond and distinguish between Single, Double, and Triple covalent bonds.

Question 2 (part 7).

Option A: Differentiate between Polar and Non-polar covalent bonds with examples.

OR

Option B: What is a Coordinate Covalent Bond? Illustrate its formation in the ammonium ion (NH_4^+).

Question 2 (part 8).

Option A: Define Hydrogen Bonding and explain its impact on the boiling point of water.

OR

Option B: Differentiate between covalent compounds and metallic substances regarding electrical conductivity.

Question 2 (part 9).

Option A: Explain why ionic compounds have high melting and boiling points compared to covalent compounds.

OR

Option B: Describe the structure and properties of metallic crystals based on the 'electron sea' model.

Question 2 (part 10).

Option A: What is the significance of Carbon-12 in establishing the relative atomic mass unit?

OR

Option B: Calculate the relative atomic mass of Carbon if it has 98.8% C-12 and 1.1% C-13.

Question 2 (part 11).

Option A: Describe the Heisenberg Uncertainty Principle and its impact on the Bohr model.

OR

Option B: Explain the dual nature of electrons as proposed by de Broglie.

Question 2 (part 12).

Option A: State the relative charges and absolute masses of a Proton, Neutron, and Electron.

OR

Option B: Explain how subatomic particles behave when passed through an electric field.

Question 2 (part 13).

Option A: Describe how radioactivity represents nuclear instability. Give one example of uranium decay.

OR

Option B: What is nuclear force, and how does it hold protons and neutrons together in the nucleus?

Question 2 (part 14).

Option A: State three uses of Diamond and relate them to its giant covalent structure.

OR

Option B: Differentiate between Intermolecular and Intramolecular forces with reference to H₂O.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Describe Rutherford's Gold Foil Experiment. Detail the observations, conclusions, and defects identified in this model of the atom. *[Marks: 6.5]*

OR

Option B: Describe Bohr's Atomic Theory in detail. State all of its core postulates and write the mathematical expression for energy differences. *[Marks: 6.5]*

Question 4.

Option A: Compare the formation, properties, and physical states of Ionic and Covalent compounds. Provide a comprehensive comparative matrix. *[Marks: 6.5]*

OR

Option B: Discuss the concept of chemical bonding. Differentiate between polar covalent, non-polar covalent, and coordinate covalent bonds, providing dot-cross structures for H_2O , CH_4 , and NH_4^+ . *[Marks: 6.5]*

Question 5.

Option A: Explain the concept of Metallic Bonding using the Electron Sea Model. Detail how this bonding accounts for electrical conductivity, thermal conductivity, and malleability. *[Marks: 6.5]*

OR

Option B: Define Isotopes. Detail the isotopic structures, abundance, and relative atomic masses of Carbon and Chlorine. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Electronic Configuration. Describe the Aufbau Principle, Pauli Exclusion Principle, and Hund's Rule. Write the electronic configuration for elements from atomic numbers 1 to 18. *[Marks: 6.5]*

OR

Option B: Discuss Intermolecular Forces. Explain Dipole-Dipole interactions and Hydrogen Bonding. Detail how hydrogen bonding affects the physical properties of water, ice, and organic polymers. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 2

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	B	Rutherford bombarded a very thin gold foil of 0.00004cm thickness with alpha particles [29].
Q2	B	Bohr stated that angular momentum of electron is an integral multiple of $h/2\pi$ [30].
Q3	C	A d-subshell can accommodate a maximum of 10 electrons [41].
Q4	C	Cobalt-60 is commonly used to irradiate cancer cells [38].
Q5	B	Proton number refers to the number of protons and is indicated by the symbol Z [33].
Q6	B	Ionic bonds are formed when one atom completely transfers electrons to another [76].
Q7	C	Atoms are electrically neutral because protons equal electrons [33].
Q8	C	An electron's relative mass is approximately 1/1836 amu [32].
Q9	B	According to Aufbau Principle, electrons fill the lowest energy 1s level first [41].
Q10	B	Sodium loses an electron to form a positively charged cation, Na ⁺ [76].
Q11	C	Ammonium ion (NH ₄ ⁺) has a coordinate covalent bond formed by donor nitrogen [84].
Q12	C	Polar covalent bonds have a significant electronegativity difference, causing partial charges [84].
Q13	A	Metallic bonds consist of a sea of delocalized valence electrons holding cations [90].
Q14	C	The standard relative atomic mass scale is based on Carbon-12 having exactly 12 amu [34].
Q15	C	These are isotopes of chlorine with 17 protons but different neutrons [37].
Q16	C	Hydrogen bonding in water causes high attractive forces and higher boiling point [86].
Q17	B	The octet rule states that atoms tend to participate in bonding to acquire eight electrons [75].

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Differentiate between Rutherford's Atomic Model and Bohr's Atomic Theory regarding electron orbits.

Model Answer A: Rutherford proposed planetary orbits with continuous energy loss. Bohr proposed fixed stationary orbits with quantized energy [29, 30].

Option B: Describe the defects of Rutherford's Atomic Model and how Bohr resolved them.

Model Answer B: Rutherford predicted continuous spectra and atomic collapse. Bohr solved this by stating electrons do not radiate energy while in fixed orbits [29, 30].

Question 2 (part 2) Solution:

Option A: Define isotopes and list the isotopes of hydrogen with their proton and neutron counts.

Model Answer A: Isotopes have same protons but different neutrons. Hydrogen: Protium (1p, 0n), Deuterium (1p, 1n), Tritium (1p, 2n) [35, 36].

Option B: Describe three practical applications of radioactive isotopes in medicine and industry.

Model Answer B: Cobalt-60 treats cancer. Iodine-131 diagnoses thyroid. Carbon-14 is used for radiocarbon dating of fossils [38].

Question 2 (part 3) Solution:

Option A: Define Aufbau Principle and write the electronic configuration of Sodium ($Z=11$) and Chlorine ($Z=17$).

Model Answer A: Aufbau principle states electrons fill lowest energy subshells first. Na: $1s^2 2s^2 2p^6 3s^1$. Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$ [41].

Option B: Differentiate between Shells and Sub-shells. List the sub-shells of the M shell.

Model Answer B: Shells are main orbits (n). Sub-shells (s, p, d, f) are sub-levels within shells. M-shell ($n=3$) has $3s, 3p, 3d$ subshells [41].

Question 2 (part 4) Solution:

Option A: Define Octet and Duplet rules. Give one example of an element obeying each.

Model Answer A: Octet: acquiring 8 electrons (e.g., Oxygen forming oxide). Duplet: acquiring 2 electrons (e.g., Lithium losing 1e to mimic Helium) [75].

Option B: Explain why noble gases are highly stable and chemically inert.

Model Answer B: Noble gases have completely filled valence s and p subshells ($ns^2 np^6$), giving them maximum electronic stability [74].

Question 2 (part 5) Solution:

Option A: Describe the formation of Cations and Anions with one chemical equation for each.

Model Answer A: Cation: $Na \rightarrow Na^+ + e^-$ (electron loss). Anion: $Cl + e^- \rightarrow Cl^-$ (electron gain) [39, 40].

Option B: Write the electronic configuration of Magnesium ion (Mg^{2+}) and Oxide ion (O^{2-}).

Model Answer B: Mg^{2+} ($Z=12$, lost 2e): $1s^2 2s^2 2p^6$. O^{2-} ($Z=8$, gained 2e): $1s^2 2s^2 2p^6$ [39, 40].

Question 2 (part 6) Solution:

Option A: Define Ionic Bond. Illustrate the formation of sodium chloride ($NaCl$) using Lewis dot structure.

Model Answer A: Ionic bond is electrostatic force between cations and anions. Na transfer electron to Cl forming Na^+ and Cl^- [76, 78].

Option B: Define Covalent Bond and distinguish between Single, Double, and Triple covalent bonds.

Model Answer B: Covalent bond involves sharing of electrons. Single (1 pair shared, H_2), Double (2 pairs, O_2), Triple (3 pairs, N_2) [80, 81].

Question 2 (part 7) Solution:

Option A: Differentiate between Polar and Non-polar covalent bonds with examples.

Model Answer A: Polar covalent has unequal sharing due to electronegativity difference (e.g., HCl). Non-polar has equal sharing (e.g., H_2) [84].

Option B: What is a Coordinate Covalent Bond? Illustrate its formation in the ammonium ion (NH_4^+).

Model Answer B: Formed when one atom provides both electrons for sharing. NH_3 (donor) shares lone pair with H^+ (acceptor) to form NH_4^+ [84].

Question 2 (part 8) Solution:

Option A: Define Hydrogen Bonding and explain its impact on the boiling point of water.

Model Answer A: Hydrogen bond is attractive force between highly electronegative atom and hydrogen. Water has strong hydrogen bonds, elevating boiling point [86].

Option B: Differentiate between covalent compounds and metallic substances regarding electrical conductivity.

Model Answer B: Covalent compounds are insulators (no free electrons). Metals are highly conductive due to free-flowing delocalized electrons [87, 91].

Question 2 (part 9) Solution:

Option A: Explain why ionic compounds have high melting and boiling points compared to covalent compounds.

Model Answer A: Ionic compounds have a rigid network held by strong electrostatic forces. Covalent compounds are held by weaker intermolecular forces [88].

Option B: Describe the structure and properties of metallic crystals based on the 'electron sea' model.

Model Answer B: Metallic crystals consist of positive metal ions surrounded by mobile delocalized valence electrons, enabling conductivity and malleability [90].

Question 2 (part 10) Solution:

Option A: What is the significance of Carbon-12 in establishing the relative atomic mass unit?

Model Answer A: Carbon-12 was chosen as the standard reference. One amu is exactly 1/12th the mass of one Carbon-12 atom [34].

Option B: Calculate the relative atomic mass of Carbon if it has 98.8% C-12 and 1.1% C-13.

Model Answer B: Relative mass = $(98.8 \times 12 + 1.1 \times 13) / 100 = 12.00$ amu [38].

Question 2 (part 11) Solution:

Option A: Describe the Heisenberg Uncertainty Principle and its impact on the Bohr model.

Model Answer A: It is impossible to simultaneously determine both the exact position and momentum of an electron, contradicting Bohr's fixed orbits [30].

Option B: Explain the dual nature of electrons as proposed by de Broglie.

Model Answer B: De Broglie proposed that moving electrons exhibit both wave-like and particle-like characteristics [31].

Question 2 (part 12) Solution:

Option A: State the relative charges and absolute masses of a Proton, Neutron, and Electron.

Model Answer A: Proton (+1, 1 amu), Neutron (0, 1 amu), Electron (-1, 1/1836 amu) [32].

Option B: Explain how subatomic particles behave when passed through an electric field.

Model Answer B: Protons deflect toward the negative plate, electrons deflect strongly toward the positive plate, neutrons pass straight [32].

Question 2 (part 13) Solution:

Option A: Describe how radioactivity represents nuclear instability. Give one example of uranium decay.

Model Answer A: Radioactivity is the spontaneous decay of unstable nuclei. U-238 decays to Thorium-234 emitting an alpha particle [34].

Option B: What is nuclear force, and how does it hold protons and neutrons together in the nucleus?

Model Answer B: Nuclear force is a strong attractive force that binds protons and neutrons together, overcoming electrostatic repulsion [31].

Question 2 (part 14) Solution:

Option A: State three uses of Diamond and relate them to its giant covalent structure.

Model Answer A: Hardness makes it useful for glass cutting, jewelry (refractive index), and medical tools [89, 90].

Option B: Differentiate between Intermolecular and Intramolecular forces with reference to H₂O.

Model Answer B: Intramolecular is covalent bond holding H and O together. Intermolecular is hydrogen bond holding H₂O molecules together [85].

Section C Model Answers

Question 3 Solution:

Option A: Describe Rutherford's Gold Foil Experiment. Detail the observations, conclusions, and defects identified in this model of the atom.

Detailed Model Answer A: Rutherford bombarded gold foil with alpha particles. Most passed undeflected (empty space); some deflected at large angles (positive nucleus) [29].

Option B: Describe Bohr's Atomic Theory in detail. State all of its core postulates and write the mathematical expression for energy differences.

Detailed Model Answer B: Bohr proposed electrons revolve in circular orbits of quantized energy. Energy is absorbed/emitted only when changing orbits: $\Delta E = E_2 - E_1 = h\nu$ [30].

Question 4 Solution:

Option A: Compare the formation, properties, and physical states of Ionic and Covalent compounds. Provide a comprehensive comparative matrix.

Detailed Model Answer A: Ionic: complete transfer, crystalline solids, high melting points, conduct in molten/aqueous state. Covalent: sharing, gases/liquids/solids, lower melting points, insulators [76, 80, 88].

Option B: Discuss the concept of chemical bonding. Differentiate between polar covalent, non-polar covalent, and coordinate covalent bonds, providing dot-cross structures for H₂O, CH₄, and NH₄⁺.

Detailed Model Answer B: Polar covalent (H₂O, unequal sharing), Non-polar covalent (CH₄, equal sharing), Coordinate covalent (NH₄⁺, one-sided donor pair) [81, 82, 84].

Question 5 Solution:

Option A: Explain the concept of Metallic Bonding using the Electron Sea Model. Detail how this bonding accounts for electrical conductivity, thermal conductivity, and malleability.

Detailed Model Answer A: Valence electrons form a mobile sea. Applying electric field drives electron flow (conductivity). Layer sliding allows bending without breaking (malleability) [90, 91].

Option B: Define Isotopes. Detail the isotopic structures, abundance, and relative atomic masses of Carbon and Chlorine.

Detailed Model Answer B: Carbon has C-12 (98.8%), C-13 (1.1%), and C-14. Chlorine has Cl-35 (75.7%) and Cl-37 (24.23%) [36, 37].

Question 6 Solution:

Option A: Explain the concept of Electronic Configuration. Describe the Aufbau Principle, Pauli Exclusion Principle, and Hund's Rule. Write the electronic configuration for elements from atomic numbers 1 to 18.

Detailed Model Answer A: Aufbau principle fills subshells in increasing energy order: 1s, 2s, 2p, 3s, 3p. Configurations of elements 1 to 18 obey these rules [41, 49].

Option B: Discuss Intermolecular Forces. Explain Dipole-Dipole interactions and Hydrogen Bonding. Detail how hydrogen bonding affects the physical properties of water, ice, and organic polymers.

Detailed Model Answer B: Dipole-dipole is force between polar molecules. Hydrogen bonding is super-strong dipole-dipole. It causes water's high boiling point and makes ice less dense than water [85, 86].

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 3

Syllabus Focus: Units 1 to 19 (Comprehensive Chemistry Evaluation)

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|-------------------|------------------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

- | | |
|---------------------|---------------|
| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

OR

Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

OR

Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

Question 2 (part 8).

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

OR

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

OR

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

OR

Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

OR

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

OR

Option B: Identify the period, group, and block of an element with atomic number 15.

Question 2 (part 13).

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 3

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Bohr model postulates that the energy of an electron in an orbit is quantized [30].
Q2	C	Fluorine has the highest electronegativity of 4.0 on the Pauling scale [60].
Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
Q5	B	Vertical columns are groups, representing families of elements with similar properties [49].
Q6	B	Transition elements are d-block elements located in groups 3 to 12, e.g., Iron [54, 64].
Q7	B	Galvanic or voltaic cells spontaneously convert chemical energy into electrical energy.
Q8	B	A neutral solution has pH exactly equal to 7 at 25 °C.
Q9	C	Carbon dioxide (CO ₂) is a major greenhouse gas absorbing infrared radiation [82].
Q10	B	Homologous series represents organic molecules sharing functional groups with regular CH ₂ increments.
Q11	C	The hydroxyl group (-OH) is the characteristic functional group of alcohols.
Q12	B	Proteins are biochemistry polymers made of monomer amino acids.
Q13	C	Distillation separates components of a mixture based on their boiling points.
Q14	B	Paper chromatography uses liquid solvent as mobile phase and paper fibers as stationary phase.
Q15	B	Magnesium has atomic number 12 and configuration 1s ² 2s ² 2p ⁶ 3s ² [50, 52].
Q16	C	Across a period, shielding effect remains nearly constant because inner shell electrons remain same [55].
Q17	C	Temporary water hardness is caused by dissolved calcium and magnesium bicarbonates.

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

Model Answer A: Galvanic cell generates electricity from chemical reaction. Electrolytic cell uses electricity to drive reaction.

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 × 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

Question 2 (part 3) Solution:

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

Option B: Describe Homologous Series and state two of its key characteristics.

Model Answer B: Homologous series share a functional group, have general formulas, and adjacent members differ by -CH₂-.

Question 2 (part 4) Solution:

Option A: What are carbohydrates? Classify them into three major classes with examples.

Model Answer A: Carbohydrates are polyhydroxy aldehydes/ketones. Classes: Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch).

Option B: Describe the structure and biological functions of DNA and RNA.

Model Answer B: DNA stores genetic code (double helix). RNA is single-stranded and directs protein synthesis.

Question 2 (part 5) Solution:

Option A: Explain the principle of Paper Chromatography.

Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

Option B: Describe simple distillation and how it separates water and ethanol.

Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

Question 2 (part 6) Solution:

Option A: Define Ionization Energy and describe its trend down a group.

Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

Option B: Define Shielding Effect and explain why it remains constant across a period.

Model Answer B: Shielding remains constant across a period because the number of inner shielding shells is constant [55].

Question 2 (part 7) Solution:

Option A: State the Octet and Duplet rules with one example of each.

Model Answer A: Octet: acquiring 8 electrons (O²⁻). Duplet: acquiring 2 electrons (Li⁺) [75, 77].

Option B: Illustrate coordinate covalent bonding in hydronium ion (H₃O⁺).

Model Answer B: Lone pair on oxygen in H₂O is donated to H⁺ forming H₃O⁺ with coordinate covalent bond [84].

Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

Model Answer A: Hard water doesn't lather with soap. Disadvantages: wastes soap, scales boilers.

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Model Answer B: Boiling decomposes soluble bicarbonates into insoluble carbonates. Clark's method adds slaked lime.

Question 2 (part 9) Solution:

Option A: Define Allotropy and name three allotropic forms of sulfur.

Model Answer A: Allotropy: existence of element in different physical forms. Sulfur allotropes: Rhombic, Monoclinic, Plastic.

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Model Answer B: Electrolysis of brine in Nelson's cell produces chlorine gas, hydrogen gas, and sodium hydroxide.

Question 2 (part 10) Solution:

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

Model Answer A: Greenhouse gases trap infrared radiation, warming the atmosphere. Excess CO₂ causes global warming [8, 82].

Option B: What is acid rain? Write the chemical equations showing its formation.

Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

Question 2 (part 11) Solution:

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers

Question 3 Solution:

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 4

Syllabus Focus: Board Syllabus Evaluation — Mock 1

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|-------------------|------------------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

- | | |
|---------------------|---------------|
| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

OR

Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

OR

Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

Question 2 (part 8).

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

OR

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

OR

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

OR

Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

OR

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

OR

Option B: Identify the period, group, and block of an element with atomic number 15.

Question 2 (part 13).

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 4

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Bohr model postulates that the energy of an electron in an orbit is quantized [30].
Q2	C	Fluorine has the highest electronegativity of 4.0 on the Pauling scale [60].
Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
Q5	B	Vertical columns are groups, representing families of elements with similar properties [49].
Q6	B	Transition elements are d-block elements located in groups 3 to 12, e.g., Iron [54, 64].
Q7	B	Galvanic or voltaic cells spontaneously convert chemical energy into electrical energy.
Q8	B	A neutral solution has pH exactly equal to 7 at 25 °C.
Q9	C	Carbon dioxide (CO ₂) is a major greenhouse gas absorbing infrared radiation [82].
Q10	B	Homologous series represents organic molecules sharing functional groups with regular CH ₂ increments.
Q11	C	The hydroxyl group (-OH) is the characteristic functional group of alcohols.
Q12	B	Proteins are biochemistry polymers made of monomer amino acids.
Q13	C	Distillation separates components of a mixture based on their boiling points.
Q14	B	Paper chromatography uses liquid solvent as mobile phase and paper fibers as stationary phase.
Q15	B	Magnesium has atomic number 12 and configuration 1s ² 2s ² 2p ⁶ 3s ² [50, 52].
Q16	C	Across a period, shielding effect remains nearly constant because inner shell electrons remain same [55].
Q17	C	Temporary water hardness is caused by dissolved calcium and magnesium bicarbonates.

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

Model Answer A: Galvanic cell generates electricity from chemical reaction. Electrolytic cell uses electricity to drive reaction.

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 × 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

Question 2 (part 3) Solution:

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

Option B: Describe Homologous Series and state two of its key characteristics.

Model Answer B: Homologous series share a functional group, have general formulas, and adjacent members differ by -CH₂-.

Question 2 (part 4) Solution:

Option A: What are carbohydrates? Classify them into three major classes with examples.

Model Answer A: Carbohydrates are polyhydroxy aldehydes/ketones. Classes: Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch).

Option B: Describe the structure and biological functions of DNA and RNA.

Model Answer B: DNA stores genetic code (double helix). RNA is single-stranded and directs protein synthesis.

Question 2 (part 5) Solution:

Option A: Explain the principle of Paper Chromatography.

Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

Option B: Describe simple distillation and how it separates water and ethanol.

Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

Question 2 (part 6) Solution:

Option A: Define Ionization Energy and describe its trend down a group.

Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

Option B: Define Shielding Effect and explain why it remains constant across a period.

Model Answer B: Shielding remains constant across a period because the number of inner shielding shells is constant [55].

Question 2 (part 7) Solution:

Option A: State the Octet and Duplet rules with one example of each.

Model Answer A: Octet: acquiring 8 electrons (O²⁻). Duplet: acquiring 2 electrons (Li⁺) [75, 77].

Option B: Illustrate coordinate covalent bonding in hydronium ion (H₃O⁺).

Model Answer B: Lone pair on oxygen in H₂O is donated to H⁺ forming H₃O⁺ with coordinate covalent bond [84].

Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

Model Answer A: Hard water doesn't lather with soap. Disadvantages: wastes soap, scales boilers.

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Model Answer B: Boiling decomposes soluble bicarbonates into insoluble carbonates. Clark's method adds slaked lime.

Question 2 (part 9) Solution:

Option A: Define Allotropy and name three allotropic forms of sulfur.

Model Answer A: Allotropy: existence of element in different physical forms. Sulfur allotropes: Rhombic, Monoclinic, Plastic.

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Model Answer B: Electrolysis of brine in Nelson's cell produces chlorine gas, hydrogen gas, and sodium hydroxide.

Question 2 (part 10) Solution:

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

Model Answer A: Greenhouse gases trap infrared radiation, warming the atmosphere. Excess CO₂ causes global warming [8, 82].

Option B: What is acid rain? Write the chemical equations showing its formation.

Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

Question 2 (part 11) Solution:

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers

Question 3 Solution:

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 5

Syllabus Focus: Board Syllabus Evaluation — Mock 2

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|----------|---------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

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| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

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Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

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Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

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Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

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Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

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Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

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Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

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Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

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Option B: Identify the period, group, and block of an element with atomic number 15.

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Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 5

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Bohr model postulates that the energy of an electron in an orbit is quantized [30].
Q2	C	Fluorine has the highest electronegativity of 4.0 on the Pauling scale [60].
Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
Q5	B	Vertical columns are groups, representing families of elements with similar properties [49].
Q6	B	Transition elements are d-block elements located in groups 3 to 12, e.g., Iron [54, 64].
Q7	B	Galvanic or voltaic cells spontaneously convert chemical energy into electrical energy.
Q8	B	A neutral solution has pH exactly equal to 7 at 25 °C.
Q9	C	Carbon dioxide (CO ₂) is a major greenhouse gas absorbing infrared radiation [82].
Q10	B	Homologous series represents organic molecules sharing functional groups with regular CH ₂ increments.
Q11	C	The hydroxyl group (-OH) is the characteristic functional group of alcohols.
Q12	B	Proteins are biochemistry polymers made of monomer amino acids.
Q13	C	Distillation separates components of a mixture based on their boiling points.
Q14	B	Paper chromatography uses liquid solvent as mobile phase and paper fibers as stationary phase.
Q15	B	Magnesium has atomic number 12 and configuration 1s ² 2s ² 2p ⁶ 3s ² [50, 52].
Q16	C	Across a period, shielding effect remains nearly constant because inner shell electrons remain same [55].
Q17	C	Temporary water hardness is caused by dissolved calcium and magnesium bicarbonates.

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

Model Answer A: Galvanic cell generates electricity from chemical reaction. Electrolytic cell uses electricity to drive reaction.

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 × 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

Question 2 (part 3) Solution:

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

Option B: Describe Homologous Series and state two of its key characteristics.

Model Answer B: Homologous series share a functional group, have general formulas, and adjacent members differ by -CH₂-.

Question 2 (part 4) Solution:

Option A: What are carbohydrates? Classify them into three major classes with examples.

Model Answer A: Carbohydrates are polyhydroxy aldehydes/ketones. Classes: Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch).

Option B: Describe the structure and biological functions of DNA and RNA.

Model Answer B: DNA stores genetic code (double helix). RNA is single-stranded and directs protein synthesis.

Question 2 (part 5) Solution:

Option A: Explain the principle of Paper Chromatography.

Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

Option B: Describe simple distillation and how it separates water and ethanol.

Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

Question 2 (part 6) Solution:

Option A: Define Ionization Energy and describe its trend down a group.

Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

Option B: Define Shielding Effect and explain why it remains constant across a period.

Model Answer B: Shielding remains constant across a period because the number of inner shielding shells is constant [55].

Question 2 (part 7) Solution:

Option A: State the Octet and Duplet rules with one example of each.

Model Answer A: Octet: acquiring 8 electrons (O²⁻). Duplet: acquiring 2 electrons (Li⁺) [75, 77].

Option B: Illustrate coordinate covalent bonding in hydronium ion (H₃O⁺).

Model Answer B: Lone pair on oxygen in H₂O is donated to H⁺ forming H₃O⁺ with coordinate covalent bond [84].

Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

Model Answer A: Hard water doesn't lather with soap. Disadvantages: wastes soap, scales boilers.

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Model Answer B: Boiling decomposes soluble bicarbonates into insoluble carbonates. Clark's method adds slaked lime.

Question 2 (part 9) Solution:

Option A: Define Allotropy and name three allotropic forms of sulfur.

Model Answer A: Allotropy: existence of element in different physical forms. Sulfur allotropes: Rhombic, Monoclinic, Plastic.

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Model Answer B: Electrolysis of brine in Nelson's cell produces chlorine gas, hydrogen gas, and sodium hydroxide.

Question 2 (part 10) Solution:

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

Model Answer A: Greenhouse gases trap infrared radiation, warming the atmosphere. Excess CO₂ causes global warming [8, 82].

Option B: What is acid rain? Write the chemical equations showing its formation.

Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

Question 2 (part 11) Solution:

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers

Question 3 Solution:

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 6

Syllabus Focus: Board Syllabus Evaluation — Mock 3

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|-------------------|------------------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

- | | |
|---------------------|---------------|
| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

OR

Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

OR

Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

Question 2 (part 8).

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

OR

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

OR

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

OR

Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

OR

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

OR

Option B: Identify the period, group, and block of an element with atomic number 15.

Question 2 (part 13).

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 6

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Bohr model postulates that the energy of an electron in an orbit is quantized [30].
Q2	C	Fluorine has the highest electronegativity of 4.0 on the Pauling scale [60].
Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
Q5	B	Vertical columns are groups, representing families of elements with similar properties [49].
Q6	B	Transition elements are d-block elements located in groups 3 to 12, e.g., Iron [54, 64].
Q7	B	Galvanic or voltaic cells spontaneously convert chemical energy into electrical energy.
Q8	B	A neutral solution has pH exactly equal to 7 at 25 °C.
Q9	C	Carbon dioxide (CO ₂) is a major greenhouse gas absorbing infrared radiation [82].
Q10	B	Homologous series represents organic molecules sharing functional groups with regular CH ₂ increments.
Q11	C	The hydroxyl group (-OH) is the characteristic functional group of alcohols.
Q12	B	Proteins are biochemistry polymers made of monomer amino acids.
Q13	C	Distillation separates components of a mixture based on their boiling points.
Q14	B	Paper chromatography uses liquid solvent as mobile phase and paper fibers as stationary phase.
Q15	B	Magnesium has atomic number 12 and configuration 1s ² 2s ² 2p ⁶ 3s ² [50, 52].
Q16	C	Across a period, shielding effect remains nearly constant because inner shell electrons remain same [55].
Q17	C	Temporary water hardness is caused by dissolved calcium and magnesium bicarbonates.

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

Model Answer A: Galvanic cell generates electricity from chemical reaction. Electrolytic cell uses electricity to drive reaction.

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 x 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

Question 2 (part 3) Solution:

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

Option B: Describe Homologous Series and state two of its key characteristics.

Model Answer B: Homologous series share a functional group, have general formulas, and adjacent members differ by -CH₂-.

Question 2 (part 4) Solution:

Option A: What are carbohydrates? Classify them into three major classes with examples.

Model Answer A: Carbohydrates are polyhydroxy aldehydes/ketones. Classes: Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch).

Option B: Describe the structure and biological functions of DNA and RNA.

Model Answer B: DNA stores genetic code (double helix). RNA is single-stranded and directs protein synthesis.

Question 2 (part 5) Solution:

Option A: Explain the principle of Paper Chromatography.

Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

Option B: Describe simple distillation and how it separates water and ethanol.

Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

Question 2 (part 6) Solution:

Option A: Define Ionization Energy and describe its trend down a group.

Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

Option B: Define Shielding Effect and explain why it remains constant across a period.

Model Answer B: Shielding remains constant across a period because the number of inner shielding shells is constant [55].

Question 2 (part 7) Solution:

Option A: State the Octet and Duplet rules with one example of each.

Model Answer A: Octet: acquiring 8 electrons (O²⁻). Duplet: acquiring 2 electrons (Li⁺) [75, 77].

Option B: Illustrate coordinate covalent bonding in hydronium ion (H₃O⁺).

Model Answer B: Lone pair on oxygen in H₂O is donated to H⁺ forming H₃O⁺ with coordinate covalent bond [84].

Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

Model Answer A: Hard water doesn't lather with soap. Disadvantages: wastes soap, scales boilers.

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Model Answer B: Boiling decomposes soluble bicarbonates into insoluble carbonates. Clark's method adds slaked lime.

Question 2 (part 9) Solution:

Option A: Define Allotropy and name three allotropic forms of sulfur.

Model Answer A: Allotropy: existence of element in different physical forms. Sulfur allotropes: Rhombic, Monoclinic, Plastic.

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Model Answer B: Electrolysis of brine in Nelson's cell produces chlorine gas, hydrogen gas, and sodium hydroxide.

Question 2 (part 10) Solution:

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

Model Answer A: Greenhouse gases trap infrared radiation, warming the atmosphere. Excess CO₂ causes global warming [8, 82].

Option B: What is acid rain? Write the chemical equations showing its formation.

Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

Question 2 (part 11) Solution:

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers**Question 3 Solution:**

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 7

Syllabus Focus: Board Syllabus Evaluation — Mock 4

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|-------------------|------------------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

- | | |
|---------------------|---------------|
| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

OR

Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

OR

Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

Question 2 (part 8).

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

OR

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

OR

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

OR

Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

OR

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

OR

Option B: Identify the period, group, and block of an element with atomic number 15.

Question 2 (part 13).

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 7

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Bohr model postulates that the energy of an electron in an orbit is quantized [30].
Q2	C	Fluorine has the highest electronegativity of 4.0 on the Pauling scale [60].
Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
Q5	B	Vertical columns are groups, representing families of elements with similar properties [49].
Q6	B	Transition elements are d-block elements located in groups 3 to 12, e.g., Iron [54, 64].
Q7	B	Galvanic or voltaic cells spontaneously convert chemical energy into electrical energy.
Q8	B	A neutral solution has pH exactly equal to 7 at 25 °C.
Q9	C	Carbon dioxide (CO ₂) is a major greenhouse gas absorbing infrared radiation [82].
Q10	B	Homologous series represents organic molecules sharing functional groups with regular CH ₂ increments.
Q11	C	The hydroxyl group (-OH) is the characteristic functional group of alcohols.
Q12	B	Proteins are biochemistry polymers made of monomer amino acids.
Q13	C	Distillation separates components of a mixture based on their boiling points.
Q14	B	Paper chromatography uses liquid solvent as mobile phase and paper fibers as stationary phase.
Q15	B	Magnesium has atomic number 12 and configuration 1s ² 2s ² 2p ⁶ 3s ² [50, 52].
Q16	C	Across a period, shielding effect remains nearly constant because inner shell electrons remain same [55].
Q17	C	Temporary water hardness is caused by dissolved calcium and magnesium bicarbonates.

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

Model Answer A: Galvanic cell generates electricity from chemical reaction. Electrolytic cell uses electricity to drive reaction.

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 × 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

Question 2 (part 3) Solution:

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

Option B: Describe Homologous Series and state two of its key characteristics.

Model Answer B: Homologous series share a functional group, have general formulas, and adjacent members differ by -CH₂-.

Question 2 (part 4) Solution:

Option A: What are carbohydrates? Classify them into three major classes with examples.

Model Answer A: Carbohydrates are polyhydroxy aldehydes/ketones. Classes: Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch).

Option B: Describe the structure and biological functions of DNA and RNA.

Model Answer B: DNA stores genetic code (double helix). RNA is single-stranded and directs protein synthesis.

Question 2 (part 5) Solution:

Option A: Explain the principle of Paper Chromatography.

Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

Option B: Describe simple distillation and how it separates water and ethanol.

Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

Question 2 (part 6) Solution:

Option A: Define Ionization Energy and describe its trend down a group.

Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

Option B: Define Shielding Effect and explain why it remains constant across a period.

Model Answer B: Shielding remains constant across a period because the number of inner shielding shells is constant [55].

Question 2 (part 7) Solution:

Option A: State the Octet and Duplet rules with one example of each.

Model Answer A: Octet: acquiring 8 electrons (O²⁻). Duplet: acquiring 2 electrons (Li⁺) [75, 77].

Option B: Illustrate coordinate covalent bonding in hydronium ion (H₃O⁺).

Model Answer B: Lone pair on oxygen in H₂O is donated to H⁺ forming H₃O⁺ with coordinate covalent bond [84].

Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

Model Answer A: Hard water doesn't lather with soap. Disadvantages: wastes soap, scales boilers.

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Model Answer B: Boiling decomposes soluble bicarbonates into insoluble carbonates. Clark's method adds slaked lime.

Question 2 (part 9) Solution:

Option A: Define Allotropy and name three allotropic forms of sulfur.

Model Answer A: Allotropy: existence of element in different physical forms. Sulfur allotropes: Rhombic, Monoclinic, Plastic.

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Model Answer B: Electrolysis of brine in Nelson's cell produces chlorine gas, hydrogen gas, and sodium hydroxide.

Question 2 (part 10) Solution:

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

Model Answer A: Greenhouse gases trap infrared radiation, warming the atmosphere. Excess CO₂ causes global warming [8, 82].

Option B: What is acid rain? Write the chemical equations showing its formation.

Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

Question 2 (part 11) Solution:

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers

Question 3 Solution:

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 8

Syllabus Focus: Board Syllabus Evaluation — Mock 5

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|-------------------|------------------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

- | | |
|---------------------|---------------|
| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

OR

Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

OR

Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

Question 2 (part 8).

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

OR

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

OR

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

OR

Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

OR

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

OR

Option B: Identify the period, group, and block of an element with atomic number 15.

Question 2 (part 13).

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 8

Section A Answer Key & Explanations

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Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
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Question 2 (part 1) Solution:

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Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 × 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

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Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

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Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

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Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

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Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

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Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

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Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

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Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

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Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers

Question 3 Solution:

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 9

Syllabus Focus: Board Syllabus Evaluation — Mock 6

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|----------|---------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

- | | |
|---------------------|---------------|
| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

OR

Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

OR

Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

Question 2 (part 8).

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

OR

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

OR

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

OR

Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

OR

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

OR

Option B: Identify the period, group, and block of an element with atomic number 15.

Question 2 (part 13).

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 9

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Bohr model postulates that the energy of an electron in an orbit is quantized [30].
Q2	C	Fluorine has the highest electronegativity of 4.0 on the Pauling scale [60].
Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
Q5	B	Vertical columns are groups, representing families of elements with similar properties [49].
Q6	B	Transition elements are d-block elements located in groups 3 to 12, e.g., Iron [54, 64].
Q7	B	Galvanic or voltaic cells spontaneously convert chemical energy into electrical energy.
Q8	B	A neutral solution has pH exactly equal to 7 at 25 °C.
Q9	C	Carbon dioxide (CO ₂) is a major greenhouse gas absorbing infrared radiation [82].
Q10	B	Homologous series represents organic molecules sharing functional groups with regular CH ₂ increments.
Q11	C	The hydroxyl group (-OH) is the characteristic functional group of alcohols.
Q12	B	Proteins are biochemistry polymers made of monomer amino acids.
Q13	C	Distillation separates components of a mixture based on their boiling points.
Q14	B	Paper chromatography uses liquid solvent as mobile phase and paper fibers as stationary phase.
Q15	B	Magnesium has atomic number 12 and configuration 1s ² 2s ² 2p ⁶ 3s ² [50, 52].
Q16	C	Across a period, shielding effect remains nearly constant because inner shell electrons remain same [55].
Q17	C	Temporary water hardness is caused by dissolved calcium and magnesium bicarbonates.

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

Model Answer A: Galvanic cell generates electricity from chemical reaction. Electrolytic cell uses electricity to drive reaction.

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 × 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

Question 2 (part 3) Solution:

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

Option B: Describe Homologous Series and state two of its key characteristics.

Model Answer B: Homologous series share a functional group, have general formulas, and adjacent members differ by -CH₂-.

Question 2 (part 4) Solution:

Option A: What are carbohydrates? Classify them into three major classes with examples.

Model Answer A: Carbohydrates are polyhydroxy aldehydes/ketones. Classes: Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch).

Option B: Describe the structure and biological functions of DNA and RNA.

Model Answer B: DNA stores genetic code (double helix). RNA is single-stranded and directs protein synthesis.

Question 2 (part 5) Solution:

Option A: Explain the principle of Paper Chromatography.

Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

Option B: Describe simple distillation and how it separates water and ethanol.

Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

Question 2 (part 6) Solution:

Option A: Define Ionization Energy and describe its trend down a group.

Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

Option B: Define Shielding Effect and explain why it remains constant across a period.

Model Answer B: Shielding remains constant across a period because the number of inner shielding shells is constant [55].

Question 2 (part 7) Solution:

Option A: State the Octet and Duplet rules with one example of each.

Model Answer A: Octet: acquiring 8 electrons (O²⁻). Duplet: acquiring 2 electrons (Li⁺) [75, 77].

Option B: Illustrate coordinate covalent bonding in hydronium ion (H₃O⁺).

Model Answer B: Lone pair on oxygen in H₂O is donated to H⁺ forming H₃O⁺ with coordinate covalent bond [84].

Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

Model Answer A: Hard water doesn't lather with soap. Disadvantages: wastes soap, scales boilers.

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Model Answer B: Boiling decomposes soluble bicarbonates into insoluble carbonates. Clark's method adds slaked lime.

Question 2 (part 9) Solution:

Option A: Define Allotropy and name three allotropic forms of sulfur.

Model Answer A: Allotropy: existence of element in different physical forms. Sulfur allotropes: Rhombic, Monoclinic, Plastic.

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Model Answer B: Electrolysis of brine in Nelson's cell produces chlorine gas, hydrogen gas, and sodium hydroxide.

Question 2 (part 10) Solution:

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

Model Answer A: Greenhouse gases trap infrared radiation, warming the atmosphere. Excess CO₂ causes global warming [8, 82].

Option B: What is acid rain? Write the chemical equations showing its formation.

Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

Question 2 (part 11) Solution:

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers

Question 3 Solution:

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .

CHEMISTRY SSC-I (GRADE 9) MODEL PAPER 10

Syllabus Focus: Board Syllabus Evaluation — Mock 7

Time Allowed: 3 Hours (Total)	Total Marks: 85 Marks
Section A: 25 Minutes (17 Marks)	Sections B & C: 2 Hours 35 Mins (68 Marks)

SECTION A (Objective Type — 17 Marks)

Instructions: Time allowed is 25 minutes. Section A is compulsory. All questions carry equal marks (1 mark each). Choose the correct option and fill in the corresponding bubble on the response sheet. Do not use pencil or overwrite.

Q1. According to Bohr's theory, the energy of an electron is:

- A) Zero
- B) Continuous
- C) Quantized
- D) Infinite

Q2. Which of the following elements has the highest electronegativity?

- A) Oxygen
- B) Nitrogen
- C) Fluorine
- D) Chlorine

Q3. The oxidation number of hydrogen in metal hydrides (like NaH) is:

- A) +1
- B) -1
- C) 0
- D) +2

Q4. The branch of chemistry that deals with nuclear reactions and radioactivity is:

- A) Industrial
- B) Nuclear chemistry
- C) Astrochemistry
- D) Biochemistry

Q5. The vertical columns in the periodic table are known as:

- A) Periods
- B) Groups
- C) Blocks
- D) Series

Q6. Which of the following is a transition element?

- A) Sodium
- B) Iron
- C) Chlorine
- D) Helium

Q7. A galvanic cell converts:

- A) Electrical to chemical energy
- B) Chemical to electrical energy
- C) Heat to work
- D) Nuclear to heat

Q8. The pH of a neutral solution at 25 degrees Celsius is:

- A) 0
- B) 7
- C) 14
- D) 1

Q9. Which gas is a major contributor to the greenhouse effect and global warming?

- | | |
|-------------------|-----------|
| A) Nitrogen | B) Oxygen |
| C) Carbon dioxide | D) Argon |

Q10. Homologous series represents organic compounds with:

- | | |
|----------------------------------|--|
| A) Same mass | B) Same functional group and regular structural increments |
| C) Different physical properties | D) No relation |

Q11. The functional group of alcohols is:

- | | |
|----------|---------|
| A) -COOH | B) -CHO |
| C) -OH | D) -CO- |

Q12. Proteins are polymers of:

- | | |
|----------------|----------------|
| A) Glucose | B) Amino acids |
| C) Fatty acids | D) Nucleotides |

Q13. Which separation technique is based on difference in boiling points?

- | | |
|-----------------|-------------------|
| A) Filtration | B) Sublimation |
| C) Distillation | D) Chromatography |

Q14. The mobile phase in paper chromatography is usually a:

- | | |
|----------|---------------------|
| A) Solid | B) Liquid (solvent) |
| C) Gas | D) Alum |

Q15. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^2$?

- | | |
|-------------|--------------|
| A) Sodium | B) Magnesium |
| C) Aluminum | D) Silicon |

Q16. The shielding effect across a period:

- | | |
|---------------------|---------------|
| A) Increases | B) Decreases |
| C) Remains constant | D) Fluctuates |

Q17. Temporary hardness of water is caused by calcium and magnesium:

- | | |
|-----------------|--------------|
| A) Sulfates | B) Chlorides |
| C) Bicarbonates | D) Nitrates |

SECTION B (Short Answers — 42 Marks)

Instructions: Answer all fourteen (14) short-answer questions. Every item has a 100% internal choice. Each question carries exactly 3 marks. Support your answers with chemical equations, diagrams, and formulas where applicable.

Question 2 (part 1).

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

OR

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Question 2 (part 2).

Option A: Define pH and calculate the pH of a 1.0×10^{-3} M HCl solution.

OR

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Question 2 (part 3).

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

OR

Option B: Describe Homologous Series and state two of its key characteristics.

Question 2 (part 4).

Option A: What are carbohydrates? Classify them into three major classes with examples.

OR

Option B: Describe the structure and biological functions of DNA and RNA.

Question 2 (part 5).

Option A: Explain the principle of Paper Chromatography.

OR

Option B: Describe simple distillation and how it separates water and ethanol.

Question 2 (part 6).

Option A: Define Ionization Energy and describe its trend down a group.

OR

Option B: Define Shielding Effect and explain why it remains constant across a period.

Question 2 (part 7).

Option A: State the Octet and Duplet rules with one example of each.

OR

Option B: Illustrate coordinate covalent bonding in hydronium ion (H_3O^+).

Question 2 (part 8).

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

OR

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Question 2 (part 9).

Option A: Define Allotropy and name three allotropic forms of sulfur.

OR

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Question 2 (part 10).

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

OR

Option B: What is acid rain? Write the chemical equations showing its formation.

Question 2 (part 11).

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

OR

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Question 2 (part 12).

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

OR

Option B: Identify the period, group, and block of an element with atomic number 15.

Question 2 (part 13).

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

OR

Option B: What is a neutralization reaction? Explain with a molecular equation.

Question 2 (part 14).

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

OR

Option B: Describe how dipole-dipole interactions occur between polar molecules.

SECTION C (Long/Extensive Answers — 26 Marks)

Instructions: Answer any four (4) extensive questions. Each question carries exactly 6.5 marks. Every item features an internal choice. Marks are divided granularly as indicated (e.g., 3.5 + 3 or 4 + 2.5 marks).

Question 3.

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations. *[Marks: 6.5]*

OR

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process. *[Marks: 6.5]*

Question 4.

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups. *[Marks: 6.5]*

OR

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations. *[Marks: 6.5]*

Question 5.

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters. *[Marks: 6.5]*

OR

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable. *[Marks: 6.5]*

Question 6.

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction. *[Marks: 6.5]*

OR

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies. *[Marks: 6.5]*

SOLUTION & GRADING GUIDE — MODEL PAPER 10

Section A Answer Key & Explanations

Q.No	Correct Option	SLO Grounding / Explanation
Q1	C	Bohr model postulates that the energy of an electron in an orbit is quantized [30].
Q2	C	Fluorine has the highest electronegativity of 4.0 on the Pauling scale [60].
Q3	B	In metal hydrides, hydrogen is highly electronegative relative to metals and has oxidation state -1.
Q4	B	Nuclear chemistry is the study of changes that occur in atomic nuclei [7].
Q5	B	Vertical columns are groups, representing families of elements with similar properties [49].
Q6	B	Transition elements are d-block elements located in groups 3 to 12, e.g., Iron [54, 64].
Q7	B	Galvanic or voltaic cells spontaneously convert chemical energy into electrical energy.
Q8	B	A neutral solution has pH exactly equal to 7 at 25 °C.
Q9	C	Carbon dioxide (CO ₂) is a major greenhouse gas absorbing infrared radiation [82].
Q10	B	Homologous series represents organic molecules sharing functional groups with regular CH ₂ increments.
Q11	C	The hydroxyl group (-OH) is the characteristic functional group of alcohols.
Q12	B	Proteins are biochemistry polymers made of monomer amino acids.
Q13	C	Distillation separates components of a mixture based on their boiling points.
Q14	B	Paper chromatography uses liquid solvent as mobile phase and paper fibers as stationary phase.
Q15	B	Magnesium has atomic number 12 and configuration 1s ² 2s ² 2p ⁶ 3s ² [50, 52].
Q16	C	Across a period, shielding effect remains nearly constant because inner shell electrons remain same [55].
Q17	C	Temporary water hardness is caused by dissolved calcium and magnesium bicarbonates.

Section B Model Answers

Question 2 (part 1) Solution:

Option A: Explain the difference between a galvanic cell and an electrolytic cell.

Model Answer A: Galvanic cell generates electricity from chemical reaction. Electrolytic cell uses electricity to drive reaction.

Option B: Define Oxidation and Reduction in terms of loss and gain of electrons.

Model Answer B: Oxidation is the loss of electrons; reduction is the gain of electrons.

Question 2 (part 2) Solution:

Option A: Define pH and calculate the pH of a 1.0 × 10⁻³ M HCl solution.

Model Answer A: pH = -log[H⁺] = -log[10⁻³] = 3.0 (Acidic solution).

Option B: What are indicators? Name two synthetic indicators and their colors in acidic media.

Model Answer B: Indicators change color with pH. Litmus (red in acid), Methyl orange (red in acid).

Question 2 (part 3) Solution:

Option A: Differentiate between alkanes, alkenes, and alkynes based on bonding.

Model Answer A: Alkanes have single bonds (C-C). Alkenes have double bonds (C=C). Alkynes have triple bonds (C≡C).

Option B: Describe Homologous Series and state two of its key characteristics.

Model Answer B: Homologous series share a functional group, have general formulas, and adjacent members differ by -CH₂-.

Question 2 (part 4) Solution:

Option A: What are carbohydrates? Classify them into three major classes with examples.

Model Answer A: Carbohydrates are polyhydroxy aldehydes/ketones. Classes: Monosaccharides (glucose), Disaccharides (sucrose), Polysaccharides (starch).

Option B: Describe the structure and biological functions of DNA and RNA.

Model Answer B: DNA stores genetic code (double helix). RNA is single-stranded and directs protein synthesis.

Question 2 (part 5) Solution:

Option A: Explain the principle of Paper Chromatography.

Model Answer A: Chromatography separates mixtures based on partition of components between stationary paper and mobile solvent.

Option B: Describe simple distillation and how it separates water and ethanol.

Model Answer B: Distillation heats a liquid mixture to vaporize the lower boiling component (ethanol), then condenses and collects it.

Question 2 (part 6) Solution:

Option A: Define Ionization Energy and describe its trend down a group.

Model Answer A: Ionization energy decreases down a group because atomic size increases, reducing nuclear hold [58].

Option B: Define Shielding Effect and explain why it remains constant across a period.

Model Answer B: Shielding remains constant across a period because the number of inner shielding shells is constant [55].

Question 2 (part 7) Solution:

Option A: State the Octet and Duplet rules with one example of each.

Model Answer A: Octet: acquiring 8 electrons (O²⁻). Duplet: acquiring 2 electrons (Li⁺) [75, 77].

Option B: Illustrate coordinate covalent bonding in hydronium ion (H₃O⁺).

Model Answer B: Lone pair on oxygen in H₂O is donated to H⁺ forming H₃O⁺ with coordinate covalent bond [84].

Question 2 (part 8) Solution:

Option A: Differentiate between hard and soft water. State two disadvantages of hard water.

Model Answer A: Hard water doesn't lather with soap. Disadvantages: wastes soap, scales boilers.

Option B: Describe how temporary hardness can be removed by boiling and Clark's method.

Model Answer B: Boiling decomposes soluble bicarbonates into insoluble carbonates. Clark's method adds slaked lime.

Question 2 (part 9) Solution:

Option A: Define Allotropy and name three allotropic forms of sulfur.

Model Answer A: Allotropy: existence of element in different physical forms. Sulfur allotropes: Rhombic, Monoclinic, Plastic.

Option B: Describe the industrial manufacture of Sodium Hydroxide using Nelson's cell.

Model Answer B: Electrolysis of brine in Nelson's cell produces chlorine gas, hydrogen gas, and sodium hydroxide.

Question 2 (part 10) Solution:

Option A: Explain the greenhouse effect and how CO₂ maintains the Earth's temperature.

Model Answer A: Greenhouse gases trap infrared radiation, warming the atmosphere. Excess CO₂ causes global warming [8, 82].

Option B: What is acid rain? Write the chemical equations showing its formation.

Model Answer B: Acid rain is rain with low pH caused by SO₂ and NO_x reacting with atmospheric moisture to form H₂SO₄ and HNO₃.

Question 2 (part 11) Solution:

Option A: Describe the role of biochemistry in understanding diseases and nutrition.

Model Answer A: Biochemistry details metabolism, diagnosing diseases via enzyme/protein tests and regulating nutrition.

Option B: What are lipids? Distinguish between fats and oils based on saturation.

Model Answer B: Lipids are water-insoluble esters of fatty acids. Fats are saturated (solid), oils are unsaturated (liquid).

Question 2 (part 12) Solution:

Option A: Write the electronic configuration of Calcium (Z=20) and Sulfur (Z=16).

Model Answer A: Ca: 1s² 2s² 2p⁶ 3s² 3p⁶ 4s². S: 1s² 2s² 2p⁶ 3s² 3p⁴ [41].

Option B: Identify the period, group, and block of an element with atomic number 15.

Model Answer B: P (Z=15) has configuration 1s² 2s² 2p⁶ 3s² 3p³. It is in Period 3, Group 15 (VA), p-block [49, 50].

Question 2 (part 13) Solution:

Option A: Define Salt. Give examples of normal, acidic, and basic salts.

Model Answer A: Salts are ionic products of acid-base neutralization. Normal (NaCl), Acidic (NaHCO₃), Basic (Pb(OH)Cl).

Option B: What is a neutralization reaction? Explain with a molecular equation.

Model Answer B: Reaction between acid and base to form salt and water, e.g., HCl + NaOH → NaCl + H₂O.

Question 2 (part 14) Solution:

Option A: Differentiate between Polar and Non-polar covalent compounds based on electronegativity.

Model Answer A: Polar covalent compounds have electronegativity differences (e.g., H-Cl). Non-polar share equally (e.g., Cl-Cl) [84].

Option B: Describe how dipole-dipole interactions occur between polar molecules.

Model Answer B: Electrostatic attraction between the partial positive end of one polar molecule and the partial negative end of another [85].

Section C Model Answers

Question 3 Solution:

Option A: Discuss Electrochemistry in detail. Explain the working and construction of a Galvanic (Voltaic) cell with a complete labeled diagram and half-cell equations.

Detailed Model Answer A: Galvanic cell: zinc anode, copper cathode. Anode: Zn → Zn²⁺ + 2e⁻. Cathode: Cu²⁺ + 2e⁻ → Cu. Generates 1.1V electrical potential.

Option B: Explain the electrolysis of molten sodium chloride in Down's cell. Provide diagrams, chemical equations, and the industrial significance of this process.

Detailed Model Answer B: Down's cell extracts sodium metal. Molten NaCl electrolyzed. Anode (carbon): 2Cl⁻ → Cl₂ + 2e⁻. Cath cathode (iron): 2Na⁺ + 2e⁻ → 2Na.

Question 4 Solution:

Option A: Define and explain Periodic Properties. Describe the periodic trends of Ionization Energy, Atomic Radius, Electron Affinity, and Electronegativity across periods and down groups.

Detailed Model Answer A: IE, EA, Electronegativity increase across a period and decrease down a group. Atomic radius decreases across a period and increases down a group [55, 56, 58, 59, 60].

Option B: Explain the modern periodic table. Discuss periods, groups, blocks (s, p, d, f), and the justification of placing elements based on electronic configurations.

Detailed Model Answer B: Elements arranged by increasing atomic number. Blocks correspond to filling subshells. Groups represent valence electrons [48, 49, 54].

Question 5 Solution:

Option A: Describe the chemical characteristics of Organic Compounds. Classify them based on functional groups. Provide a table of functional groups for Alcohols, Aldehydes, Ketones, Carboxylic acids, and Esters.

Detailed Model Answer A: Organic compounds are carbon-based. Alcohols (-OH), Aldehydes (-CHO), Ketones (-CO-), Carboxylic Acids (-COOH), Esters (-COOR).

Option B: Explain water pollution. Detail three major water pollutants (industrial, agricultural, domestic) and outline water treatment procedures to make it drinkable.

Detailed Model Answer B: Water pollutants: heavy metals, fertilizers/pesticides, domestic sewage. Treatment: screening, aeration, coagulation, sedimentation, filtration, disinfection.

Question 6 Solution:

Option A: Explain the concept of Chemical Equilibrium. Detail the Law of Mass Action and derive the equilibrium constant expression (K_c) for a general reversible reaction.

Detailed Model Answer A: Law of mass action states rate of reaction is proportional to active masses. For $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

Option B: Discuss the layers of the atmosphere. Explain the chemistry of ozone depletion, the role of chlorofluorocarbons (CFCs), and mitigation strategies.

Detailed Model Answer B: Atmosphere: Troposphere, Stratosphere, Mesosphere, Thermosphere. CFCs release chlorine free radicals under UV, decomposing O_3 to O_2 .
